

MIHC: Multi-View Interpretable Hypergraph Neural Networks with Information Bottleneck for Chip Congestion Prediction

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1. Overview

- First interpretable framework for chip congestion prediction using multi-view hypergraph learning
- Balanced fusion of netlist (topological) and layout (geometric) information through net-based bridging
- Novel Information Bottleneck mechanism identifies critical congestion-correlated regions for interpretability
- Superior performance: Up to **16.67%** NMAE reduction on ISPD2015, **5.26%** on CircuitNet-N28, with robust cross-design generalization

2. Problem Setup

Circuit Congestion: Occurs when routing resources are exceeded by wires, impacting reliability. Given multi-view data: grid layout $P \in \mathbb{R}^{M \times N \times K}$ and cell hypergraph $G = (V, E)$.

Goal: Predict $\hat{C}^G \in \mathbb{R}^{M \times N}$ and $\hat{C}^C \in \mathbb{R}^{|V|}$

$$\mathcal{L}_{\text{sup}} = \frac{1}{M \times N} \sum_{i,j} |C^G[i,j] - \hat{C}^G[i,j]|^2 + \frac{1}{|V|} \sum_k |C^C[k] - \hat{C}^C[k]|^2$$

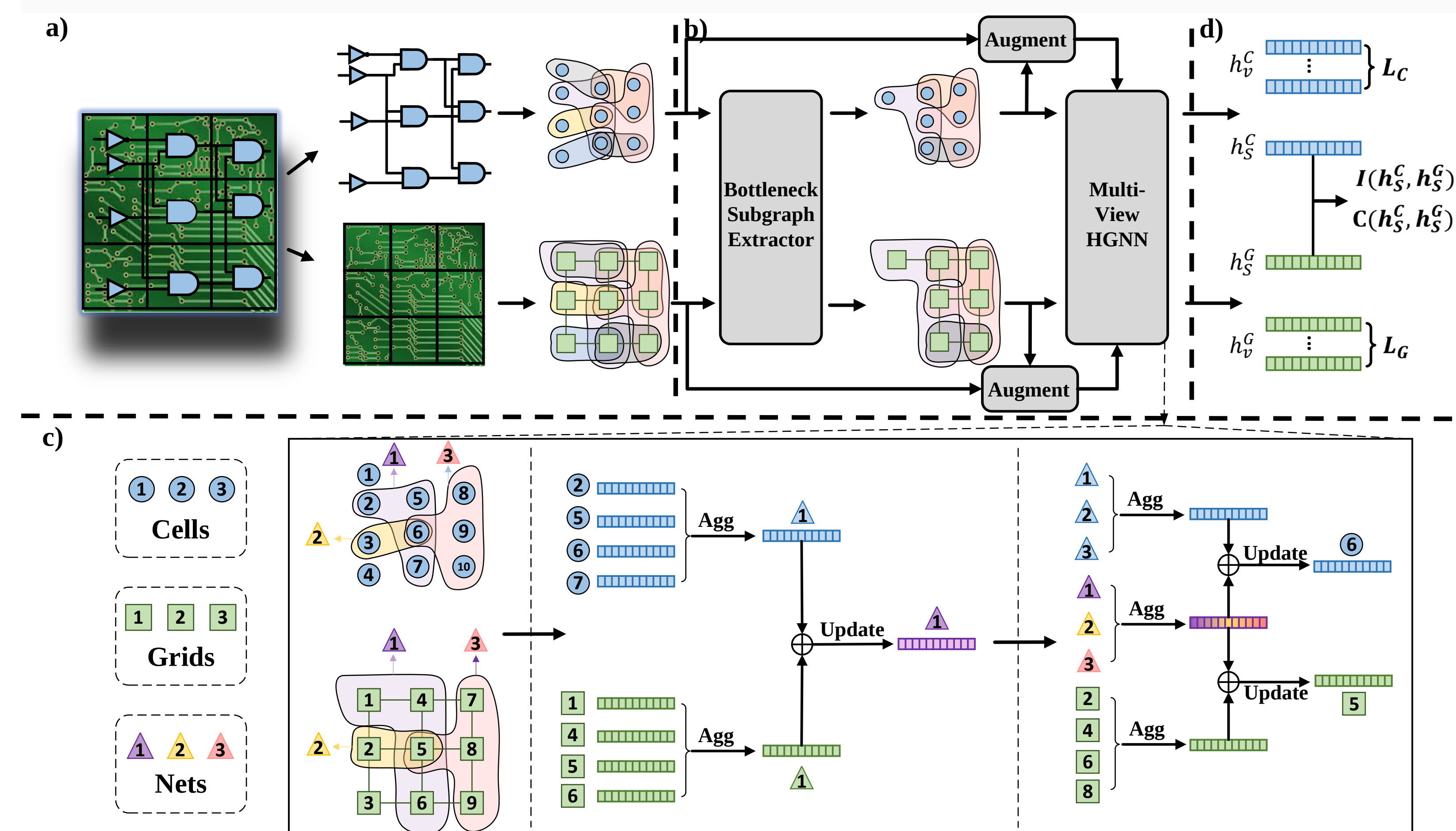
Challenges:

- Single-view methods miss complementary information
- Multi-view methods favor one data type
- Black-box predictions lack interpretability

3. Contributions

- Multi-View Representation Learning:** Balanced HGNN fusing netlist and layout via net-based hyperedge bridging
- Interpretable Bottleneck:** First Information Bottleneck for circuit congestion, identifying critical regions
- Dual-View Prediction:** Unified framework for both cell-based and grid-based predictions
- SOTA Performance:** Significant improvements with robust cross-design generalization

4. MIHC Framework



(a) Process Module: Convert multi-view data into hypergraphs

- Cell-based $H^C = (V^C, E^C)$: topological; Grid-based $H^G = (V^G, E^G)$: geometric
- Both hyperedges built through nets $\rightarrow$ balanced fusion

(b) Interpretable Subgraph Module: Multi-View Information Bottleneck

Extract bottleneck subgraphs H_S^C and H_S^G :

$$\max_{H_S^C} I(H^G; H_S^C) - \beta^C I(H^C; H_S^C | H^G)$$

- HNN extracts: $H_S^C = (\mathbf{M}^C, \mathbf{X}^C \odot \mathbf{p})$ where $\mathbf{p}$ is node probability
- Feature augmentation with bottleneck for focused learning

(c) Multi-View Representation Learning: MV-HGNN

Node-to-Hyperedge aggregation and fusion:

$$h_e^N = \sigma(W_e^N[h_e^C || h_e^G])$$

Hyperedge-to-Node update with both view-specific and fused info:

$$h_v^{C'} = \sigma(W_v^{C'}[\hat{h}_v^C || \hat{h}_v^N]), \quad h_v^{G'} = \sigma(W_v^{G'}[\hat{h}_v^G || \hat{h}_v^N])$$

(d) Prediction: $\hat{y}_v^C = f^C(h_v^{C'})$, $\hat{y}_v^G = f^G(h_v^{G'})$ $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cont}} + \mathcal{L}_{\text{IB}} + \mathcal{L}_{\text{sup}}$, where $\mathcal{L}_{\text{cont}}$ aligns multi-view representations, $\mathcal{L}_{\text{IB}}$ extracts critical subgraphs, and $\mathcal{L}_{\text{sup}}$ minimizes prediction error.

5. Experiments

1. Placement-level Results (intra-design)

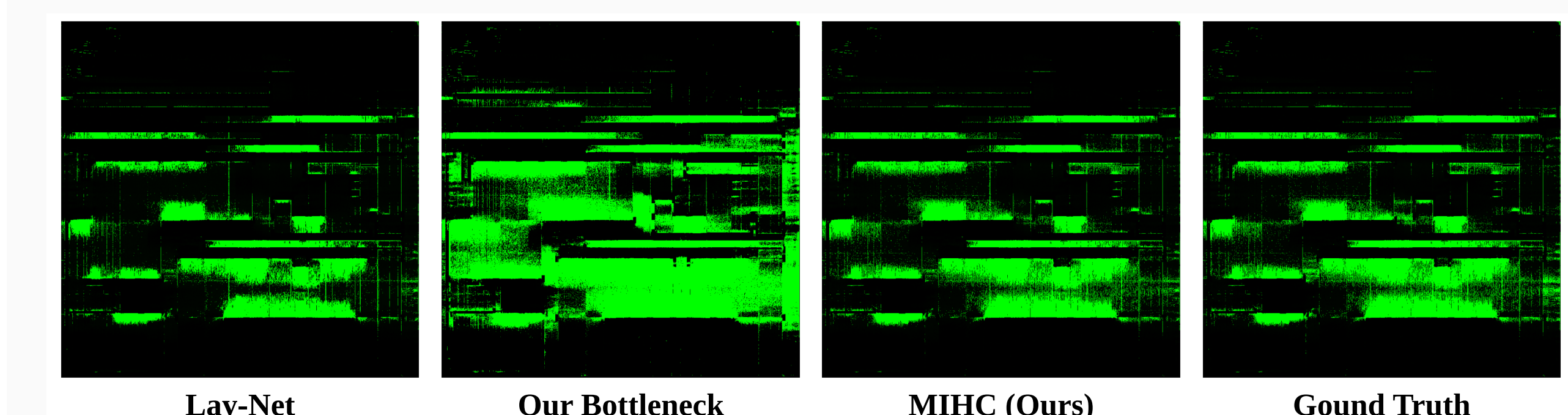
Model	ISPD2015-B			ISPD2015-F			CircuitNet-N28		
	NMAE↓	Pearson↑	Spearman↑	NMAE↓	Pearson↑	Spearman↑	NMAE↓	Pearson↑	Spearman↑
<i>Cell-based View</i>									
CircuitGNN	0.034	0.598	0.611	0.051	0.344	0.336	0.040	0.609	0.500
LHNN	0.033	0.629	0.627	0.043	0.448	0.384	0.039	0.617	0.509
DE-HNN	0.032	0.645	0.632	0.042	0.467	0.395	0.038	0.638	0.511
MIHC	0.029	0.687	0.689	0.035	0.524	0.512	0.036	0.661	0.522
<i>Grid-based View</i>									
CircuitGNN	0.037	0.561	0.201	0.042	0.364	0.162	0.047	0.547	0.508
LHNN	0.033	0.651	0.252	0.037	0.464	0.221	0.043	0.646	0.533
RouteNet	0.038	0.522	0.198	0.042	0.362	0.161	0.047	0.541	0.501
CircuitFormer	0.034	0.649	0.231	0.038	0.453	0.196	0.044	0.622	0.526
Lay-Net	0.031	0.667	0.301	0.035	0.484	0.256	0.041	0.671	0.558
MIHC	0.030	0.675	0.297	0.032	0.503	0.271	0.040	0.682	0.572

2. Design-level Results (cross-design)

Direction Model	ISPD2015-B				ISPD2015-F			
	<i>Cell-based View</i>		<i>Grid-based View</i>		<i>Cell-based View</i>		<i>Grid-based View</i>	
A→B	CircuitGNN	0.058	0.376	0.061	0.366	0.071	0.343	0.331
	LHNN	0.054	0.398	0.057	0.389	0.067	0.368	0.349
	DE-HNN/Lay-Net	0.051	0.415	0.053	0.411	0.063	0.385	0.394
	MIHC	0.048	0.448	0.051	0.436	0.059	0.414	0.405
B→A	CircuitGNN	0.063	0.351	0.064	0.348	0.075	0.337	0.318
	LHNN	0.058	0.379	0.060	0.369	0.070	0.354	0.337
	DE-HNN/Lay-Net	0.053	0.405	0.055	0.392	0.065	0.378	0.367
	MIHC	0.049	0.428	0.054	0.415	0.062	0.399	0.388

Models trained on design group A/B and tested on entirely unseen design group B/A to evaluate **cross-design generalization**.

3. Interpretability Visualization



Our bottleneck subgraph successfully identifies critical congestion regions, enabling **focused prediction** closely matching ground truth.

6. Conclusion

- Net-based bridging enables balanced multi-view fusion
- Consistent SOTA across datasets in both cell and grid views
- Strong generalization on entirely unseen circuit designs
- Bottleneck subgraphs reveal critical structures causing congestion